

VERMILION AGDM, AB, Canada

WMO: 712780

Lat: 53.3439N

Lon: 110.8819W

Elev: 623

StdP: 94.06

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-31.8	-28.6	-35.5	0.1	-31.8	-32.0	0.2	-28.4	9.9	-6.1	8.7	-8.9	1.8	150	0.662

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.7	28.7	17.8	26.8	16.9	25.1	16.3	19.7	26.0	18.5	24.5	17.3	23.2	3.8	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.3	13.4	23.2	16.1	12.3	21.6	15.0	11.5	20.1	59.0	26.3	54.9	24.3	51.2	23.4	23.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.2	8.1	7.1	-36.4	31.7	2.7	1.7	-38.3	32.9	-39.9	34.0	-41.4	34.9	-43.4	36.2		
			-36.4	22.1	2.6	1.2	-38.3	22.9	-39.9	23.6	-41.3	24.3	-43.2	25.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.5	-12.2	-12.4	-5.3	3.6	10.4	14.7	17.1	15.6	10.8	3.7	-5.5	-11.9	(6)
(7)		DBStd	12.38	8.32	7.86	7.27	5.30	4.33	3.07	2.74	3.20	4.42	4.86	7.03	7.66	(7)
(8)		HDD10.0	3429	687	626	473	200	48	4	0	2	42	202	466	679	(8)
(9)		HDD18.3	5828	946	859	732	443	246	114	57	95	228	454	716	937	(9)
(10)		CDD10.0	678	0	0	0	6	61	144	221	174	66	6	0	0	(10)
(11)		CDD18.3	36	0	0	0	0	1	5	19	9	1	0	0	0	(11)
(12)		CDH23.3	791	0	0	0	2	59	112	305	217	93	3	0	0	(12)
(13)		CDH26.7	174	0	0	0	0	8	18	74	46	27	0	0	0	(13)
(14)	Wind	WSAvg	3.2	3.2	3.0	3.4	3.8	3.7	3.1	2.7	2.6	2.9	3.4	3.3	2.9	(14)
(15)	Precipitation	PrecAvg	416	18	13	16	29	43	73	79	58	35	18	16	19	(15)
(16)		PrecMax	605	42	27	44	91	101	132	166	148	97	38	54	33	(16)
(17)		PrecMin	223	5	4	4	2	6	15	23	9	2	2	2	7	(17)
(18)		PrecStd	88	10	6	9	21	26	30	38	25	24	10	12	7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.6	5.5	12.1	22.3	27.7	28.7	31.3	30.2	30.0	22.8	12.7	6.0	(19)
(20)			MCWB	2.3	2.3	6.1	11.0	13.7	17.1	19.7	17.9	16.2	12.8	6.3	2.6	(20)
(21)		2%	DB	3.6	3.2	8.0	19.0	25.0	26.3	28.9	27.7	26.0	17.3	8.2	3.3	(21)
(22)			MCWB	1.3	0.8	3.8	9.1	12.8	16.0	19.5	17.4	14.9	9.7	4.2	0.9	(22)
(23)		5%	DB	1.9	1.1	5.5	15.7	22.4	24.3	26.5	25.8	22.7	14.3	5.2	1.2	(23)
(24)			MCWB	0.2	-0.4	2.4	7.5	11.8	15.1	18.3	17.1	13.7	8.0	2.2	-0.7	(24)
(25)		10%	DB	-0.3	-0.7	3.7	12.7	20.0	22.3	24.7	23.7	19.5	11.6	3.0	-1.1	(25)
(26)			MCWB	-1.7	-2.2	1.3	6.2	10.6	14.2	17.5	16.4	12.2	6.8	1.0	-2.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.6	2.7	6.8	11.7	15.1	19.0	22.0	20.5	17.1	13.0	6.8	3.0	(27)
(28)			MCDWB	5.0	5.1	11.2	20.7	23.9	26.1	28.6	26.4	27.5	21.9	11.6	5.9	(28)
(29)		2%	WB	1.5	1.1	4.2	9.6	13.6	17.4	20.3	19.1	15.5	10.7	4.4	1.1	(29)
(30)			MCDWB	3.4	2.8	7.7	18.3	22.4	23.9	26.7	25.1	24.5	15.7	7.7	3.1	(30)
(31)		5%	WB	0.2	-0.3	2.6	7.8	12.5	16.2	19.2	17.8	14.1	8.8	2.6	-0.6	(31)
(32)			MCDWB	1.9	1.2	5.2	14.5	20.6	22.2	25.3	24.0	21.6	13.5	4.7	1.1	(32)
(33)		10%	WB	-1.7	-2.2	1.4	6.5	11.4	15.1	18.0	16.8	12.7	7.1	1.1	-2.4	(33)
(34)			MCDWB	-0.3	-0.8	3.5	12.1	18.9	20.6	23.6	22.8	18.5	11.3	2.9	-1.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.1	10.9	10.3	11.7	13.8	12.0	12.7	13.4	13.5	11.4	9.2	9.4	(35)
(36)			MCDBR	10.5	12.1	11.9	17.3	17.7	15.9	16.5	17.7	19.6	16.6	11.9	11.5	(36)
(37)			MCWBR	8.7	10.1	8.4	9.0	8.0	7.3	8.4	8.7	9.5	9.7	8.2	9.5	(37)
(38)		5% WB	MCDWB	10.4	11.8	11.3	16.1	15.8	13.7	14.8	15.4	17.9	15.7	10.9	11.3	(38)
(39)			MCWBR	8.8	9.9	8.1	8.9	7.5	7.1	8.1	8.3	9.1	9.6	7.8	9.6	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.247	0.293	0.287	0.331	0.351	0.372	0.366	0.378	0.313	0.277	0.250	0.220	(40)
(41)		taud		2.323	2.202	2.379	2.362	2.352	2.326	2.368	2.312	2.472	2.545	2.448	2.359	(41)
(42)		Ebn at Noon		756	802	908	902	896	877	877	841	869	841	759	735	(42)
(43)		Edh at Noon		56	89	92	108	115	119	113	113	84	63	50	44	(43)
(44)	All-Sky Solar Radiation	RadAvg		0.99	2.05	3.56	4.70	5.83	5.95	6.17	5.03	3.54	2.06	0.99	0.74	(44)
(45)		RadStd		0.10	0.15	0.28	0.41	0.41	0.43	0.33	0.38	0.38	0.29	0.11	0.07	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (65 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page